

Four passive changes can cut modeled vibration by ≥30%

Gulo 3000 | Experimental model + design simulations | Powertrain Dynamics Group | ME 495

First prototype to investigate: a shaft-path damper that preserves modeled steady-speed response.

1 Measure and model the shaft

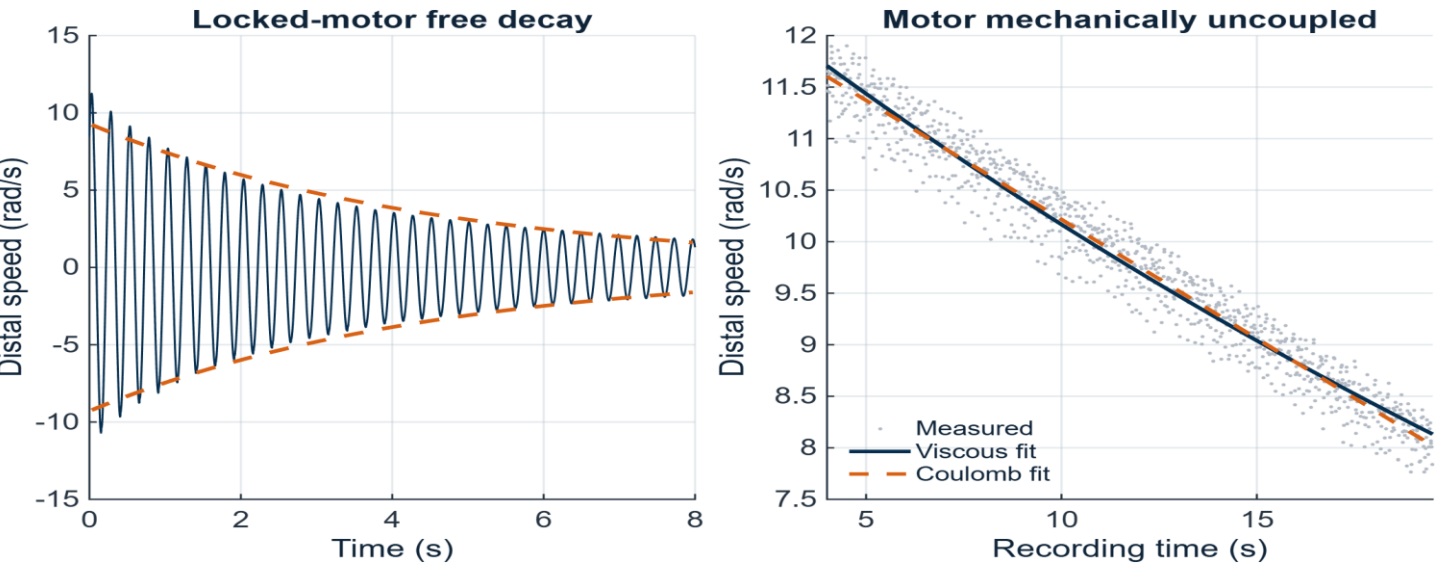
Motor switching excites torsional vibration.
Goal: validate its source and identify four ≥30% solutions.

PHOTO TO TAKE — physical test rig
Take a clear front/oblique photo. Label the motor, flywheels 1 and 2, shaft, distal bearing and both tachometers.

CAD / SCHEMATIC TO ADD — two-inertia system
Capture a simplified CAD assembly or draw a clean schematic. Label J_c , J_f , K , B_s , B_m , B_b , voltage input V_{in} and speed output ω_2 .

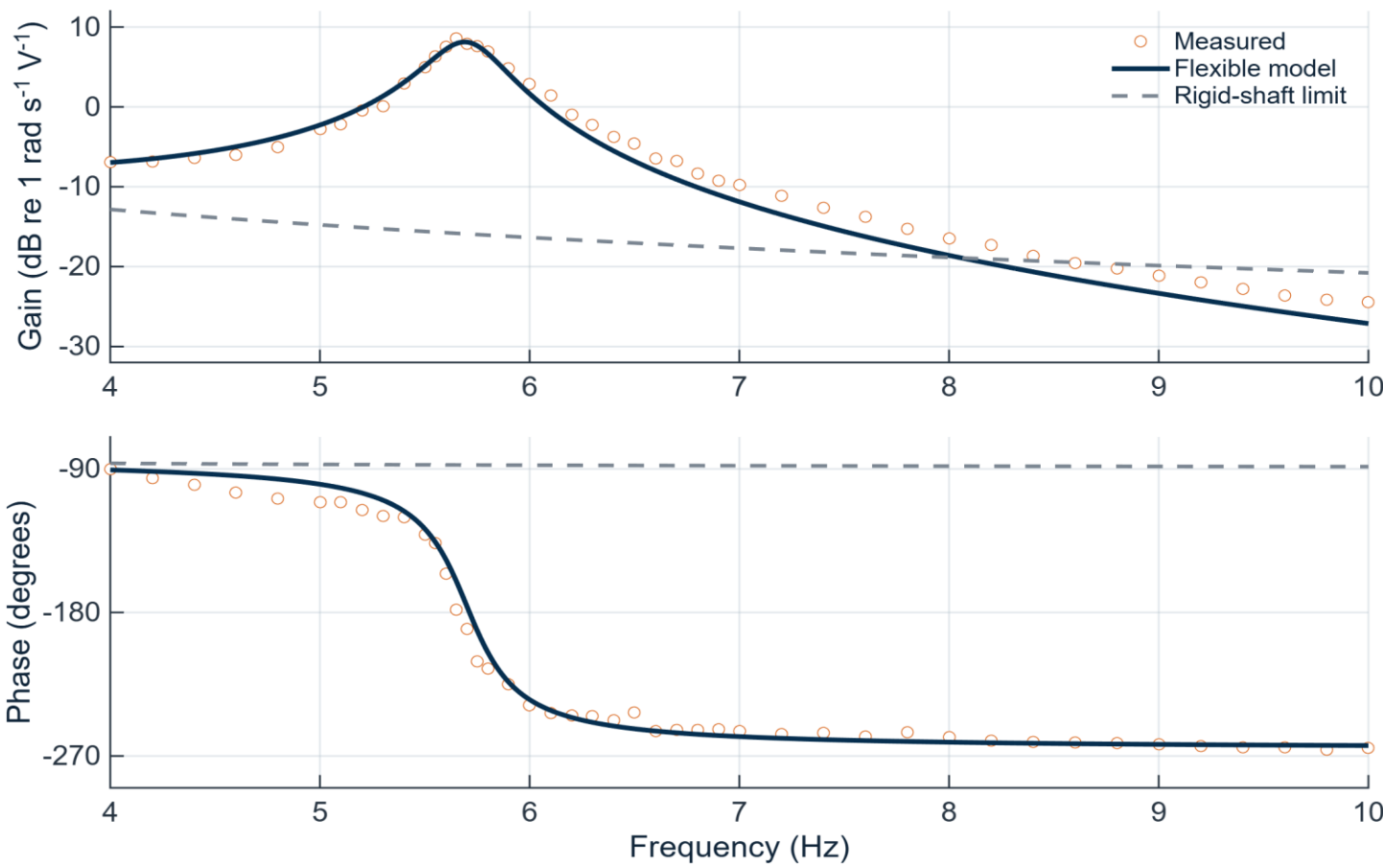
Parameters come from measurements and mechanics

Parameter	Identified value
Motor resistance R_m	2.495 Ω
Motor constant K_m	0.0982 V·s/rad
Combined damping B_{tr} / B_{tl}	0.00143 / 0.00781 N·m·s/rad
Effective shaft stiffness K	2.203 N·m/rad (refined)



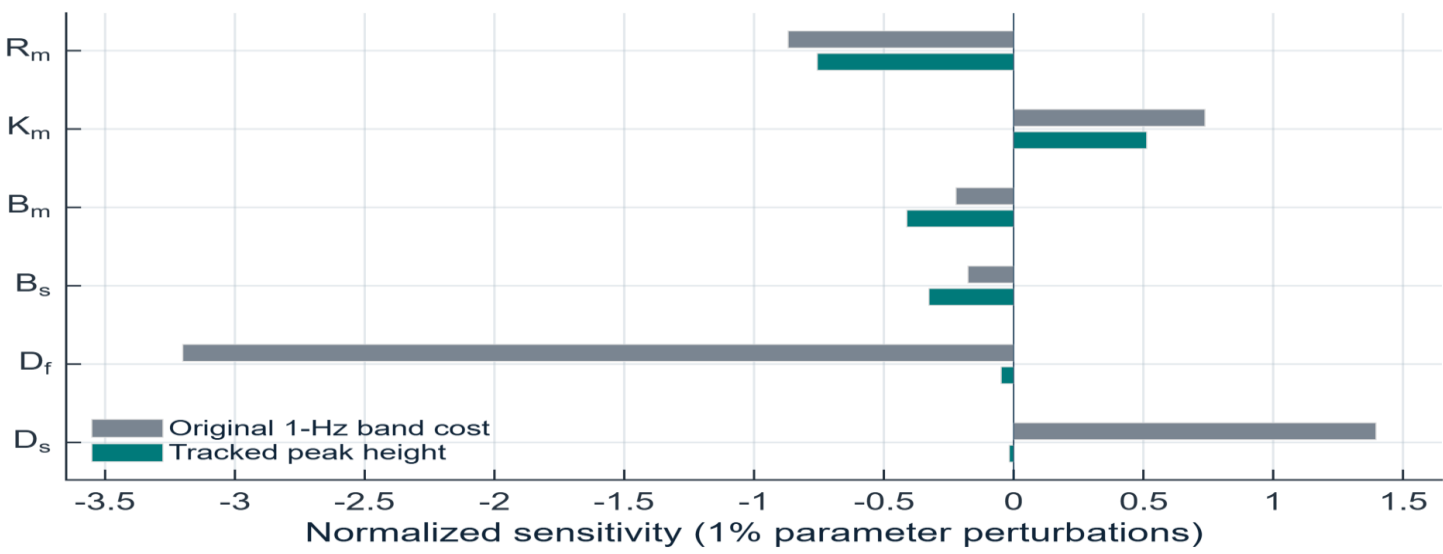
3 sweeps + 28 time records. Free decay identifies damping; the uncoupled-motor test estimates external loss.

2 The flexible model explains resonance



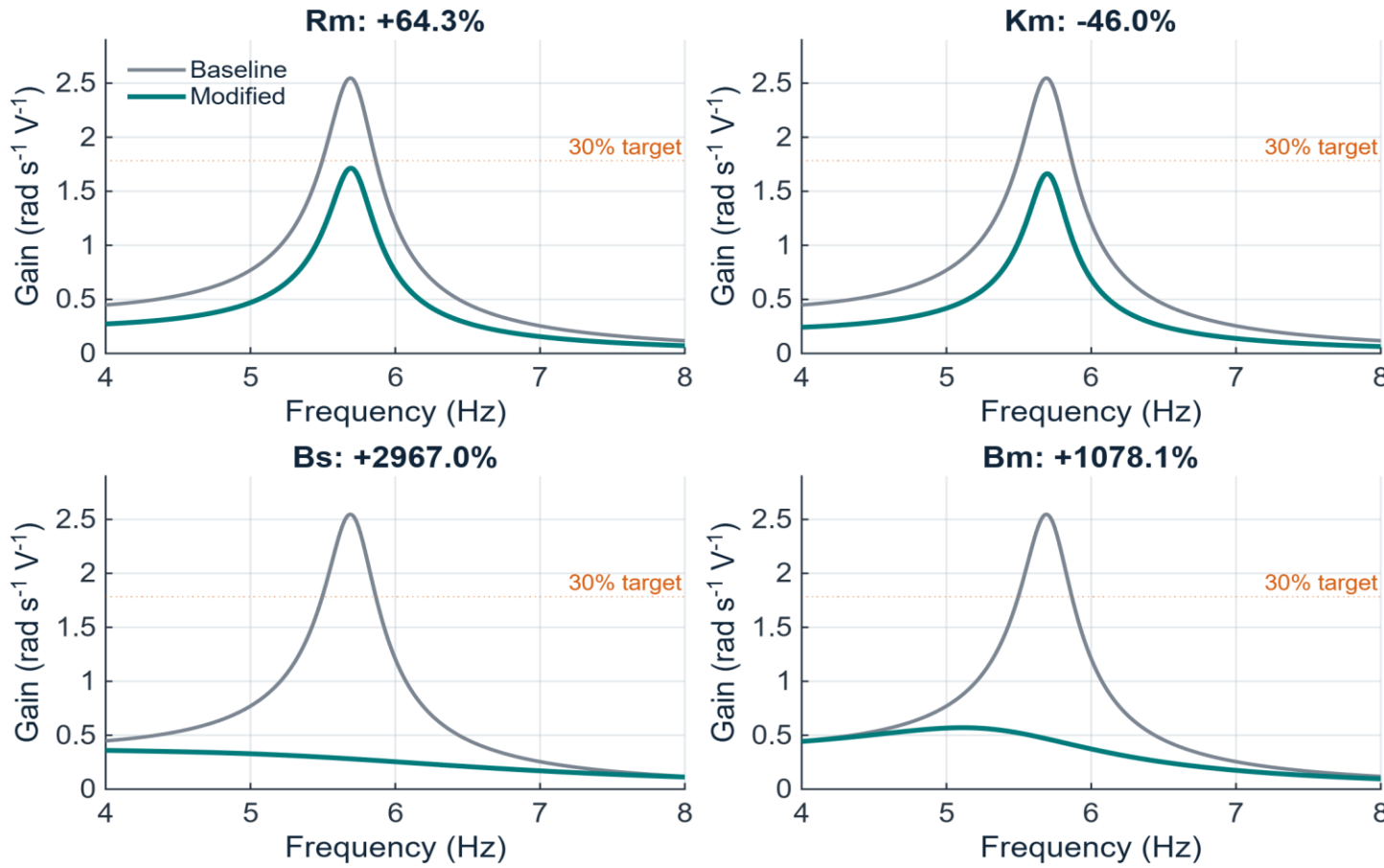
Resonance: 5.65 Hz measured / 5.69 Hz model
4–10 Hz fit error: 1.65 dB magnitude; 7.02° phase

Sensitivity separates attenuation from detuning



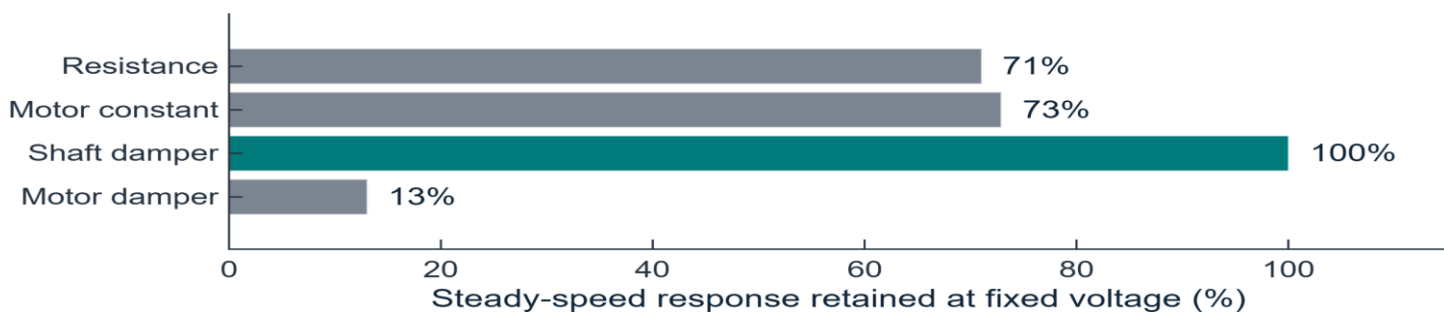
All 12 parameters were screened. Diameter strongly shifts the peak; R_m , K_m , B_m and B_s most strongly change its height.

3 Four quantified design alternatives



Change	Target value	Gain ↓	Pulse ↓
Resistance	4.10 Ω	32.7%	38.5%
Motor constant	0.053 V·s/rad	34.7%	45.0%
Shaft damping	0.039 N·m·s/rad	85.9%	32.2%
Motor damping	0.077 N·m·s/rad	77.6%	32.2%

Shaft damper: no local peak; gain assessed over 4–10 Hz.
All four reduce the 1-Hz response cost by ≥30%.



Prioritize the shaft damper. Its large damping increase needs hardware testing; the other options sacrifice steady-speed response.